

Protocol



Easy and accurate protein structure prediction using ColabFold

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Abstract

Since its public release in 2021, AlphaFold2 (AF2) has made investigating biological questions, by using predicted protein structures of single monomers or full complexes, a common practice. ColabFold-AF2 is an open-source Jupyter Notebook inside Google Colaboratory and a command-line tool that makes it easy to use AF2 while exposing its advanced options. ColabFold-AF2 shortens turnaround times of experiments because of its optimized usage of AF2's models. In this protocol, we guide the reader through ColabFold best practices by using three scenarios: (i) monomer prediction, (ii) complex prediction and (iii) conformation sampling. The first two scenarios cover classic static structure prediction and are demonstrated on the human glycosylphosphatidylinositol transamidase protein. The third scenario demonstrates an alternative use case of the AF2 models by predicting two conformations of the human alanine serine transporter 2. Users can run the protocol without computational expertise via Google Colaboratory or in a command-line environment for advanced users. Using Google Colaboratory, it takes <2 h to run each procedure. The data and code for this protocol are available at <https://protocol.colabfold.com>.

Key points

- We present an outline of how to use ColabFold to perform structure prediction of monomers, complexes and alternative conformations and guidance on interpreting the results through appropriate confidence metrics and visualizations.

- Integrating MMseqs2's quick homology search, ColabFold enables accelerated structure prediction compared with AlphaFold2 at similar accuracy, while exposing many advanced parameters. ColabFold can be accessed through a Google Colaboratory notebook for beginners and a command-line interface for advanced users.

Key references

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Introduction

Protein prediction models and ColabFold

Predicting the 3D structure of a protein from its sequence alone has long been a formidable task in the field of structural biology. The progress of machine learning models has made significant strides toward achieving this goal. AlphaFold2 (AF2)¹ and RoseTTAFold^{2,3} represent these groundbreaking models. For the first time, they offer computational methods capable of producing protein structure predictions nearly indistinguishable from experimentally solved structures, given sufficient sequence information. Specifically, AF2 is an end-to-end neural network composed of two main modules. The first module processes the information about the input amino-acid (AA) sequence (query) and generates hypotheses about which AAs are in contact with one another on the basis of a multiple sequence alignment (MSA). The second module aggregates these hypotheses to predict the structure (i.e., the 3D coordinates for each of the atoms). Two of the key ideas behind the AF2 network are the use of the deep-learning attention mechanism⁴, which allows the network to better identify AAs in contact, and refinement of the prediction by passing it through the modules several times. Similar principles guided the design of RoseTTAFold, resulting in a different network architecture whose exceptional accuracy is comparable to that of AF2.

AF2 was first designed to predict the structure of a single chain, a single protein chain. However, its models have been successfully used to predict interactions between multiple chains or complexes^{5–7}. In addition, the AF2 model has been further developed and trained specifically on multimeric input, resulting in AlphaFold-multimer⁸. Since then, many other models have been developed that use AF2 or its concepts as a foundation^{9–16}.

ColabFold⁷ is an integrated protein prediction solution aimed at simplifying the process of structure modeling for the user. As such, it offers both an easy interface to various protein prediction models and pre- and post-processing procedures. ColabFold has two interfaces: web-based, using Google Colaboratory notebooks (in short and hereafter: Colab), and command-line tools. Its web-based interface includes five notebooks: AlphaFold2.ipynb (for using either AF2 or AF2-multimer), RoseTTAFold2.ipynb, RoseTTAFold.ipynb (mainly for legacy), ESMFold.ipynb and OmegaFold.ipynb (briefly described below in Alternatives to ColabFold-AF2). The web-based interface requires free registration to Colab and is primarily designed for making a single or a small-batch prediction. The command-line interface includes only the AF2 and AF2-multimer prediction models and allows for batch predictions by processing multiple input sequences. Because the AF2 models are among the most accurate currently published ones, this protocol is focused on AlphaFold2.ipynb and the command-line interface, which are jointly denoted as ‘ColabFold-AF2’.

A key computation step in protein structure prediction with AF2 models is that of an MSA, a collection of sequences that share some degree of similarity (homology) with the input query in a way that provides information on their evolutionary relationships at the residue level, facilitating the identification of conserved regions and co-evolutionary patterns across different species. ColabFold uses a custom MMseqs2^{17,18} homology search server for fast and sensitive MSA generation, which is 40–60 times faster than other tools⁷, facilitating the prediction of hundreds of structures a day. Using the ProteinGym and CASP15 benchmarks, it has been shown that increasing the number of homologs in the MSA and their diversity by searching against large environmental databases improves prediction accuracy^{19,20}. Therefore, by default, the ColabFold MSA server uses two databases, UniRef²¹ and ColabFoldDB⁷, a large environmental database of >700 million sequences constructed from various metagenomic resources. Because of this publicly available MSA server, ColabFold users avoid the need for storing such large databases locally on their computer.

AF2 structure prediction can be improved by providing templates, which are known structures of proteins that have sequence similarity to the query. To detect templates, ColabFold first computes a smaller MSA for the query by using only UniRef homologs and from that MSA produces a position-specific scoring matrix. This position-specific scoring matrix can be thought of as a generalization of the query and is used to search the Protein Data

Bank (PDB)²², a communal resource for solved structures. Although templates generally do not significantly enhance the accuracy of AF2 in the case of MSAs with sufficient homologs^{19,23,24}, they contribute valuable information when the MSA is lacking. Removing MSA information in targeted areas allows for the integration of template data to model a specific conformation for certain residues or the entire protein. In addition to processing the MSA and templates, the behavior of protein prediction models is controlled by various parameters, such as the number of times the prediction will be iterated through the network (number of recycles). ColabFold sets these parameters to meaningful default values, saving the need to modify them in the usual case, while offering an easy way to do so, if desired. This optimizes the usage of AF2 models for the user, allowing for efficient experimental turnaround times. Upon completion of the prediction process, ColabFold produces various visualizations of the predicted structure²⁵ and the prediction quality.

Because of its simplicity and functionality, ColabFold has been widely adopted in numerous studies, and its public MSA server is used tens of thousands of times a day. Its applicability spans many biological fields. For example, it has been used to solve the structure of two members of the central AvrE family of bacterial effectors²⁶, revealing their β -barrel structure. This prediction prompted a series of cryo-EM imaging and other experiments, eventually leading to the full characterization of the biochemical function of this effector family. In another study²⁷, scientists used ColabFold-AF2 to predict the structures of dozens of nucleoporins, the building blocks of the nuclear pore complex. The resulting models were extremely accurate and were thus used for integrative modeling, covering >90% of the human nuclear pore complex. This protocol guides the reader on how to use ColabFold-AF2 for solving similar biological questions.

Development of the protocol

To demonstrate the classic use of ColabFold-AF2 for predicting static protein structures, we focus on the human glycosylphosphatidylinositol transamidase (GPIT) protein. The structure of the GPIT complex, which catalyzes the attachment of glycosylphosphatidylinositol to the endoplasmic reticulum membrane, has recently been determined experimentally by using single-particle cryo-EM at a resolution of 3.1 Å²⁸. The procedure presented in this protocol results in a predicted GPIT structure, which is remarkably similar to the experimentally solved one, with a root mean square deviation of 1.87 Å, meaning that the distance between the two protein models is very low, indicating great structural similarity.

A non-classic use case of ColabFold-AF2 may interest readers who investigate proteins that shift between conformations, such as transporters and receptors. Although AF2 was designed for predicting static structures, del Alamo et al.²⁹ have recently shown that certain manipulations to its input can, in some cases, tweak it to predict a landscape of structures spanning several conformations. At the heart of this approach lies the following logic. Proteins that undergo conformational changes are likely to have AA pairs, which are in strong structural interaction in one conformation but exhibit different interaction in another. These AA pairs will tend to co-evolve, meaning that when examined over many homologs from various species, a change in one AA will often be associated with a change in the other AA of the pair. Thus, when an MSA with many homologs ('deep MSA') is provided to the AF2 models, these AA pairs carry a signal that prompts the AF2 models to predict the conformation in which they interact. However, when the MSA depth is reduced by removing homologs, the co-evolution signal is weakened, potentially allowing the AF2 models to predict alternative conformations.

Furthermore, additional strategies can contribute to conformational sampling by increasing the uncertainty of the network (i.e., having less certainty in a single dominant conformation^{30,31}). These factors include providing templates of alternative conformations or activating dropout layers in the AF2 neural network. Dropout layers are used during the neural network training and prompt the network to produce alternative solutions for the same input by randomly deactivating certain weights. Thus, enabling these layers during inference enhances the variety of predicted structures. Using AF2 to accurately sample the conformational space is under ongoing research, leading to the development of new methods^{32,33}, as well as to the characterization of current limitations^{34,35}.

Protocol

In this protocol, we modify and extend the protocol proposed by del Alamo et al.²⁹ and demonstrate this ability by using the human alanine serine transporter 2 (ASCT2), a Na⁺-independent neutral AA transporter. This transporter is a homotrimer, which has at least two conformations, depending on whether it is facing the extracellular (outward)³⁶ or intracellular (inward) side³⁷. Through tweaking of ColabFold-AF2, we show the capacity to predict these diverse structural states.

Overview of the protocol

This paper outlines a comprehensive protocol (Fig. 1) for using ColabFold-AF2 for monomer (Procedures 1 and 2) and complex (Procedures 3 and 4) predictions and for an ad-hoc approach for conformational sampling (Procedures 5 and 6). Procedures 1, 3 and 5 are web-based approaches using Colab, and Procedures 2, 4 and 6 are local approaches using a command-line interface. Using the command-line version of the protocol requires basic familiarity with the Unix/Linux shell and a workstation capable of handling AF2 models. We demonstrate Procedures 1–4 by using the human GPIT protein. Procedures 1 and 2 predict each of its five subunits, FIGU, FIGK, FIGT, FIGS and GPAA1, as a monomer, and Procedures 3 and 4 predict them jointly as a complex. In Procedures 5 and 6, we use the human ASCT2. Readers can follow the detailed cell-by-cell instructions for each procedure or refer to Box 1 and Fig. 2 for a quick start. ColabFold-AF2 has >15 tunable parameters. These should not overwhelm first-time users, because most are expected to be kept at their default values. In Box 2, we describe the 10 parameters that users are most likely to tune as they gain experience. For clarity, we divide the described parameters into categories by their role. However, these parameters are spread over several of the notebook's cells. In each procedure, we instruct the user on how to provide the input protein sequence to ColabFold, and in Box 3, we give full details concerning ColabFold's accepted input and output formats. Box 4 provides information on the various

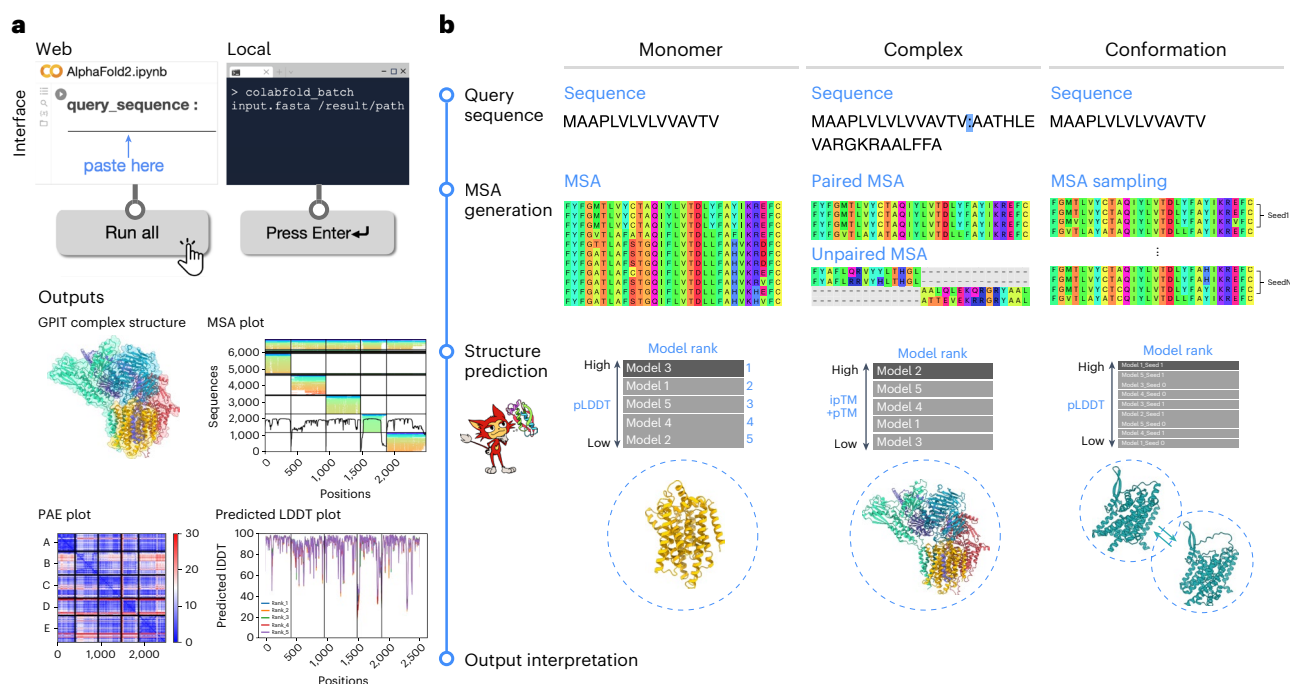


Fig. 1 | Protocol overview. **a**, ColabFold-AF2 allows initiating protein prediction either through a single click (web) or command (local). As a result, ColabFold provides the predicted structure along with plots for confidence scores (PAE and pLDDT; see Box 4 and Anticipated results) and MSA coverage and diversity. **b**, This protocol details monomer, complex and conformation predictions. Upon receiving an input protein sequence, ColabFold-AF2 begins MSA generation with MMseqs2 and proceeds to structure prediction using

AF2 models. This step generates five structure models, which are iteratively refined and ranked by their confidence scores (pLDDT and ipTM; see Box 4). In this step, strategies like MSA depth reduction can induce predictions with various conformations. The five predicted structures are saved in PDB/JSON files (Box 3). ipTM, interface predicated template modeling; PAE, predicted aligned error; pLDDT, predicted local distance difference test; pTM, predicted template modeling.

BOX 1

Quick start

ColabFold-AF2 allows protein prediction through a single click, referred to as 'quick start'. For a quick start, open a web browser and navigate to <https://alphafold.colabfold.com>. Next, paste the AA sequence of a protein of interest into the `query_sequence` field and click Runtime → Run all in the menu bar (Fig. 2). Of note, in the case of running several predictions consecutively, click Restart session and run all instead of Run all, to avoid any conflicts or memory leakage. This sequentially executes every cell

in the notebook without a break, and the currently running cell is indicated by a spinning circle on the left side. By default, ColabFold computes five structure models. After each model is generated, the predicted structures and result plots are displayed. Once all cells are processed, a pop-up window will appear, offering to download the various result files (see Box 3 for details about output formats) as a single compressed (zip) file. Please note that the runtime of these steps can vary depending on the GPU assigned by Colab.

confidence measures computed by ColabFold-AF2, and Box 5 is intended for users who do not wish to use the default ColabFold server for MSA computation. The Anticipated results section of this protocol contains general explanations on interpreting ColabFold's plots and output, followed by a demonstration of the interpretation process for each of the procedures' examples. This protocol is primarily designed for biologists aiming to conduct structural analysis and does not require coding expertise.

Alternatives to ColabFold-AF2

The first tool, which used the AF2 models, was a command-line interface by the AlphaFold team at Google DeepMind (<https://github.com/google-deepmind/alphafold>; see <https://www.ebi.ac.uk/training/online/courses/alphafold/> for a tutorial). ColabFold followed, offering the first Colab-based notebook. The debut of ColabFold was then followed by a dedicated AF2 notebook and AF2-multimer notebook⁸. Other alternatives that use the basic AF2 architecture include

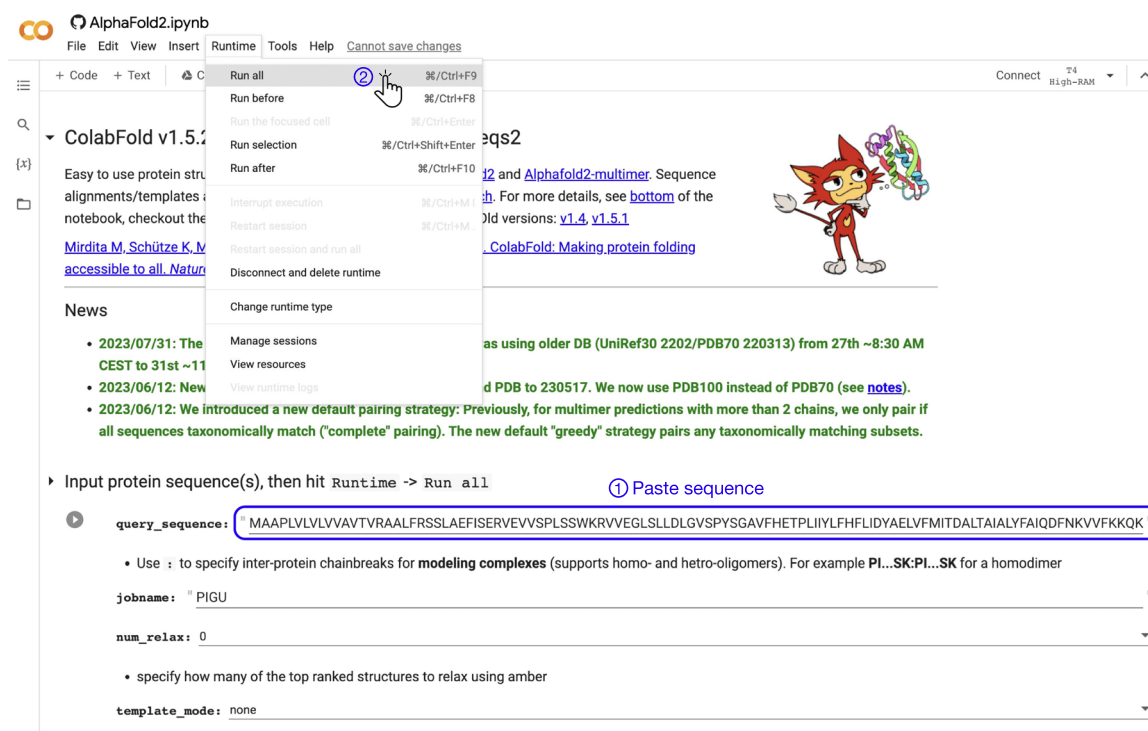


Fig. 2 | Quick start. Paste the query sequence (1) and execute all prediction steps at once by clicking Runtime → Run all (2).

BOX 2

Parameter configuration

Tuning homology search: MSA-generation options

- **msa_mode**
 - Use the ColabFold MSA server: search against the UniRef database only (`mmseqs2_uniref`) or UniRef and ColabFoldDB (`mmseqs2_uniref_env`, default).
 - **custom**: upload a pre-computed MSA.
 - **single_sequence**: predict by using a single sequence without an MSA. This option is useful for de novo–designed proteins with no homology to natural proteins.
- **pair_mode**: For heteromeric complex predictions, ColabFold first generates MSAs separately for each chain. Then, in a process called ‘pairing’, it identifies and marks sequences from the same (sub-)species across all MSAs, to enhance prediction quality.
 - **paired**: perform pairing and retain only sequences that can be paired.
 - **unpaired**: skip pairing.
 - **unpaired_paired** (default): pair and retain all sequences.
- **pairing_strategy**: This parameter takes effect when pairing is enabled.
 - **complete**: consider a sequence as ‘paired’ if it has a taxonomic match in every chain (all MSAs).
 - **greedy** (default): a sequence is paired if it has a taxonomic match in at least two chains (two MSAs).

Searching for alternative conformations: MSA-sampling options

- **max_msa**: controls the number of sequences used for structure prediction and is provided as `max_msa_clusters`: `max_extra_msa`.
 - **max_msa_clusters**: the number of randomly chosen MSA sequences, which serve as center points in a clustering procedure preceding prediction with the AF2 network.
 - **max_extra_msa**: MSA sequences that were not selected as cluster centers and are used for extra computation by the AF2 network.

`max_msa_clusters` corresponds to N_{seq} , and `max_extra_msa` corresponds to N_{extra_seq} , as defined in the Suppl. section 1.1 of Jumper et al.¹. By default (auto), these values are set to 512;5120. Lower values are useful when searching for alternative conformations (explained in Procedures 5 and 6).

- **num_seeds**: indicates the number of different random seeds to use. By default (1), a single seed value will be used. These seeds determine random components throughout prediction (e.g., when selecting the sequences, which serve as cluster centers). Setting a higher value can increase the chance of obtaining a better confidence score in cases in which the MSA is very small and templates are lacking. In combination with `max_msa`, it can help explore alternative conformations.
- **use_dropout**: activates dropout layers during inference, which prompts the AF2 neural network to be less confident in a single conformation.

Improving or reproducing results: model prediction options

- **model_type**: specifies the prediction model for monomers (`alphafold2_ptm`, `deepfold_v1`) and complexes (`alphafold2_multimer_v1`, `_2`, `_3`). By default (auto), the best models, `alphafold2_ptm` and `alphafold_multimer_v3`, will be used. Using other models is mainly used for reproducing older results (but see also Troubleshooting).
- **template_mode**: specifies whether to incorporate templates (`pdb100`, default) or not (`none`) or use custom templates (`custom`).
- **num_recycles**: controls the number of times a prediction is re-fed to the model. By default, it is set to 3, except for `model_type=alphafold2_multimer_v3`, for which it is set to 20.
- **recycle_early_stop_tolerance** (in short `tol`): controls the convergence criterion, which is considered together with `num_recycles`. By default, it is set to 0.0, except for `model_type=alphafold2_multimer_v3`, for which it is set to 0.5.

software reimplementations like FastFold¹⁰ and OpenFold¹², which offer adaptations to fit heterogeneous compute clusters and the ability to retrain the network from ground up with novel weights, respectively. Recently, DeepFold¹⁶ modified the AF2 pipeline, focusing on backbones and side-chain prediction, resulting in an alternative set of weights available through ColabFold (Box 2). Uni-fold¹³ is an alternative for AlphaFold-multimer, with optimized model performance on a graphics processing unit (GPU), a retrained model and novel weight sets.

Other tools, such as OmegaFold³⁸, ESMfold¹⁵, RGN2³⁹, trRosettaX-single⁴⁰ and Helixfold-single¹¹ offer an alternative to the AF2 architecture that bypasses the need to compute MSAs by using pre-trained language models to process a single protein sequence. Although these language model–based structure predictors are faster than MSA-based structure predictors, they are generally also less accurate^{15,41}.

ColabFold’s alternative-to-AF2 notebooks, ESMFold.ipynb, OmegaFold.ipynb and RoseTTAFold2.ipynb, are especially useful for fast monomer prediction (ESMfold) and large complex prediction (RoseTTAFold2³). However, to date, the AF2 models remain among the most accurate published models⁴¹.

BOX 3

Input and output formats

When starting from a single protein, ColabFold accepts the sequence in either FASTA or CSV/TSV. If the user provides a pre-computed MSA, ColabFold requires the A3M format, which is also the output format in cases in which the MSA is not pre-computed. ColabFold can operate on multiple input files by taking in a list of sequences in FASTA/CSV format or a directory containing FASTA/A3M files.

FASTA: Each entry has a header starting with '>', followed by sequence lines. Example:

```
>PIGU_trim
MAAPLVLVVAVTVRAALFRSSLAEFISERVE
>PIGT_trim
ARDSLREELVITPLPSGDVAATFQFRTRWDSELQREGVSHY
```

CSV: The first line is always 'id,sequence'. Each line that follows contains the header and the sequence, separated by a comma. Examples:

Monomer	Complex
id,sequence	id,sequence
PIGU_trim,MAAPLVLVLV	GPIT_trim,MAAPLVLVLV:ARDSLREELV
PIGT_trim,ARDSLREELV	

A3M begins with a super-header, followed by the aligned sequence entries. Each entry has a header starting with '>', followed by sequence lines.

- The super-header, starting with '#', contains two comma-separated lists separated by a tab. The first list indicates the length of each chain, while the second list indicates its cardinality. Each chain appears only once, regardless of its cardinality.
- The MSA is described with respect to the first (query) sequence. An AA that is aligned to an AA in the query is shown in upper case, an insertion relative to the query is shown in lower case and a deletion is shown as '-'. Example A3M for different input types:

Monomer	Hetero-oligomer
#28 1	#28,20 4,2
>example_monomer_query	>example_hetero6mer_A4B2_query
MAAPLVLVVAVTVRAALFRSSLAEFI	MAAPLVLVVAVTVRAALFRSSLAEFITAVNYPFVDTMDKFDKITK
>found_homolog_1	>found_homolog_paired1
MAFPLALVLVAVTVR-ALFRSSLAEFI	--FPLALVLVAVTVRAALFRSSLAEFITTLVNYPFVDTMDKFDITF
>found_homolog_2	>found_homolog_unpaired_chain1
...	-AFPLALVLVAVTVRAALFRSSLAEFI-----
	>found_homolog_unpaired_chain2
	-----TTLVNYPFVDTMDKFDKITF
	...

Output: ColabFold computes five protein models and reports them in three types of files:

- PDB-format for reporting the structures. See <https://pdb101.rcsb.org/learn/guide-to-understanding-pdb-data/introduction> for details. Reading the values stored in PDB files is possible by using text editors and visualizing the structure by using tools like ChimeraX⁵⁰ and PyMOL (<https://github.com/schrodinger/pymol-open-source>). Within Colab, the predicted PDB files are visualized by using 3Dmol.js²⁵.
- JSON hierarchical data format for reporting the models' confidence measures and the configuration of the parameters used for prediction. Reading JSON files is possible through text editors or web browsers like Firefox, among others.
- PNG image format for visualizing the alignment coverage and confidence scores.

Limitations of the protocol and software

As the size of the input grows, the execution time and GPU random access memory (RAM) usage of the AF2 models increase. Given Colab's capabilities, the maximum protein sequence size that can be computed varies depending on the allocated GPU. The T4 GPU with 16 GB of

BOX 4

Prediction confidence measures

AF2 has its own set of confidence measures, corresponding to traditional measures for structure prediction, whose computation requires knowledge of the true protein structure. Because the true structure is generally unknown, AF2, through training on a set of proteins with known structures, has learned to predict these confidence measures alongside predicting the structure itself. These measures are named ‘pX’, where X is a traditional measure and p stands for ‘predicted’.

predicted local distance difference test (pLDDT): For every AA in the query, AF2 predicts its LDDT-Ca score⁵¹ on a scale of 0 (bad) to 100 (excellent). Each AA’s α carbon can be described by its distances to neighboring α carbons (e.g., within a radius of 15 Å) in the true structure. The superposition-free LDDT-Ca metric reflects how preserved these distances are in the prediction model, in which the confidence for each score range is as follows:

- >90: highly confident
- 70 < pLDDT < 90: confident in a backbone level
- 50 < pLDDT < 70: lower confidence, interpret cautiously
- <50: potentially disordered regions; should be ignored

Predicted aligned error (PAE): For every amino-acid x in the query, AF2 predicts a set of scores on a scale of 0 (excellent) to >30 (bad). Each score reflects the expected displacement of x in angstroms in the predicted structure relative to the unknown true structure, when the two structures are aligned on some other AA y. PAE is

especially useful when evaluating the prediction of multi-domain/chain proteins, for which the placement of each domain is important. Specifically, if AAs that are part of one domain score well also when the alignment point is outside their domain, it suggests inter-domain confidence.

predicted template modeling (pTM) score: the TM score⁵² is expressed by using the distance between the predicted position of each AA and its true position, when the predicted and true structures are optimally aligned. AF2 predicts an approximate TM score by replacing the optimal alignment, which is infeasible to compute, with the row in the PAE matrix (i.e., an alignment on a single AA) where the total error is lowest (for full details, see Suppl. pages 37 and 38 of Jumper et al.¹). The computed pTM scores range from 0 (bad) to 1 (excellent), often interpreted as:

- >0.8: highly confident with congruent topology and backbone
- 0.5 or 0.7 < pTM < 0.8: reliable fold for single- or multi-domain proteins, respectively
- <0.2: no correlation with the true structure, intrinsically disordered protein or no MSA/templates

interface pTM (ipTM): For complexes, AlphaFold-multimer predicts a modified pTM score, ipTM⁸, which takes into account only the inter-subunit distances, estimating the prediction accuracy of interfaces. Similar to pTM, ipTM ranges from 0 (bad) to 1 (excellent), where an ipTM score > 0.85⁵³ is considered reliable.

GPU RAM, available for all Colab users, handles ~1,500 AAs, and the A100 with 40 GB of GPU RAM, available in Colab Pro+, can process up to 3,300 AAs in a single run. For longer proteins, users are encouraged to use their own local or cloud-based GPU resources or consider running predictions with RoseTTAFold2.ipynb. In addition, ColabFold permits exchanging more system RAM for longer runtimes, which may be an advantageous trade-off for specific

BOX 5

Custom MSA generation

Instead of using the default MSA server, users can provide pre-computed MSAs in A3M format (Box 3). The following are alternatives for MSA generation.

- Local colabfold_search: compute an MSA locally with the command:

```
$ colabfold_search input.fasta database/msas
```

The input file (input.fasta in this example) should contain the query sequence(s), and database/ should indicate a path to the sequence database. The execution will generate A3M-formatted MSAs in the msas folder.

- Application programming interface (API) hosting: For greater flexibility, users can also host their own application programming

interface and pass its address to --host-url when running colabfold_batch.

```
$ colabfold_batch input.fasta /path/to/results
--host-url "https://api.example.org"
```

Online tools: Users can use tools, such as the HHblits Toolkit server^{54,55} (<https://toolkit.tuebingen.mpg.de/tools/hhblits>) for web-based MSA generation. Select the tool’s settings for achieving maximal sensitivity. In the case of HHblits, we recommend a protocol similar to RoseTTAFold²: using three iterations, setting the result list size to 10,000 and applying a strict E-value cut-off of 10^{-20} . If no or only a few homologs are found, relax the E-value threshold.

use cases. Furthermore, ColabFold developers do not have control over the Google-Colab service, which means specific details, from the appearance of menu bars to allocated GPUs, may change because of updates in Google-Colab. Finally, despite its accuracy and in contrast to experimental validation, ColabFold, as any other computational method, provides only predictions about the protein structure. Such predictions can deviate from the real structure, even if they are scored as highly confident.

Experimental design

One of the advantages of using ColabFold-AF2 is that it usually does not require prior knowledge beyond the AA sequence of the protein of interest. However, some readers may find it helpful to use various bioinformatic tools for exploring their protein in advance. This exploration may include (but is not limited to) querying the PDB with the protein sequence to search for similar proteins whose structures have been solved by using tools like InterPro⁴² to identify domains and other important sites in the protein or tools like ADOPT⁴³ to identify disordered regions. After such an exploration, the reader can decide, for example, to trim off disordered regions before structure prediction, and in any case, the reader will be able to interpret the ColabFold-AF2 results in light of the exploration's main findings. Furthermore, in the case of searching for alternative conformations (Procedures 5 and 6), similar conformations can be provided as custom templates to ColabFold-AF2 (Box 2).

By default, ColabFold-AF2 produces five protein models, which are in essence structural hypotheses with confidence scores. We recommend that users examine and compare all five models because agreement between them strengthens the evidence in the predicted structure, whereas differences can reveal areas that require more attention, need to be disregarded or distinguish between alternative conformations.

Materials

Equipment

Protein sequence queries used for demonstrating the procedures

- PIGU: <https://protocol.colabfold.com/PIGU.fasta>
- PIGK: <https://protocol.colabfold.com/PIGK.fasta>
- PIGT: <https://protocol.colabfold.com/PIGT.fasta>
- PIGS: <https://protocol.colabfold.com/PIGS.fasta>
- GPAA1: <https://protocol.colabfold.com/GPAA1.fasta>
- GPIT complex: <https://rcsb.org/fasta/entry/7W72>
- ASCT2: <https://rcsb.org/fasta/entry/7BCQ>

ColabFold software

- Web-based ColabFold-AF2 notebook: <https://alphafold.colabfold.com>
- Code repository and alternative notebooks: <https://github.com/sokrypton/ColabFold>

Hardware

Running the web-based version requires a Google account (free). Throughout this protocol, unless stated otherwise, the freely available T4 GPU is used. To reproduce the examples by using command-line instructions, we recommend a Linux workstation with 16 GB of RAM and Compute Unified Device Architecture (CUDA)-capable GPU of the Volta generation or newer with ≥16 GB of GPU RAM. See the Troubleshooting section for older GPUs.

Equipment setup

Web-based ColabFold within Colab

Colab is a proprietary web-based environment offered by Google to host Jupyter notebooks with a series of code blocks called 'cells'. Colab provides logged-in users with free CPU and GPU resources for software execution. Sign in to Google and navigate to <https://alphafold.colabfold.com>.

Protocol

Downloading and installing ColabFold locally

Local installation of ColabFold requires the installer script LocalColabFold (available together with a comprehensive set-up guide at <https://github.com/YoshitakaMo/localcolabfold>). LocalColabFold is compatible with various operating systems, including Windows Subsystem for Linux 2 (freely available for Windows 10 or later), Linux and macOS (albeit without GPU support). It uses the Python package managers pip and Conda (available at <https://docs.conda.io>). Run the following commands to install ColabFold locally on a Linux system. Throughout this protocol, lines starting with \$ refer to instructions that should be typed in the command-line interface.

```
$ wget https://raw.githubusercontent.com/YoshitakaMo/localcolabfold/main/install_colabbatch_linux.sh
$ bash install_colabbatch_linux.sh
$ export PATH="/path/to/your/localcolabfold/colabfold-conda/bin:$PATH"
```

ColabFold database download

Typically, command-line users are not required to download large homology databases, because ColabFold automatically connects to an online server containing such databases for MSA generation. Because this server is a shared resource among ColabFold users, an Internet protocol (IP)-address-based rate-limiting access is implemented to ensure fairness. Thus, users who wish to compute many hundreds of structures or set up their own local server are referred to the script `setup_databases.sh`, which downloads the required databases for local MSA generation. To run this script, MMseqs2 (15-6f452 or later, available at <https://mmseqs.com>) is required. Databases of protein sequences and templates are also available at <https://colabfold.mmseqs.com>. Obtain the script and grant it permissions:

```
$ wget https://raw.githubusercontent.com/sokrypton/ColabFold/main/setup_databases.sh
$ chmod +x setup_databases.sh
```

Before executing the script, please refer to the GitHub README section at <https://github.com/sokrypton/ColabFold#generating-msas-for-large-scale-structurecomplex-predictions> for execution instructions, which ensure optimal performance for large databases, such as those used for protein structure prediction.

Procedure 1: monomer prediction (web)

▲ **CRITICAL** In the following procedure, we predict each of the five monomers of the GPIT complex individually by using ColabFold-AF2 within Colab. We explicitly guide the reader through predicting the structure of the PIGU subunit, and the other four subunits can be predicted analogously. Here, we provide step-by-step instructions for each cell (Fig. 3), which can be run one after the other from top to bottom. Users may also refer to Box 1 and Fig. 2 for a quick start.

Predicting the structure of PIGU

● TIMING ~20 min (T4 GPU)

1. Open a web browser and navigate to <https://alphafold.colabfold.com>. GPU usage is enabled by default, which can be verified by navigating to Runtime → Change runtime type in the menu bar.

◆ TROUBLESHOOTING

Protocol

- Paste the AA sequence of PIGU into the `query_sequence` field. However, now run only this single cell by clicking on the triangle icon on the left.

◆ TROUBLESHOOTING

▲ **CRITICAL STEP** To obtain the relevant sequences, see the Equipment section of this protocol.

The screenshot displays a Google Colab notebook titled 'PIGU.ipynb' with a URL of 'https://colabfold.com'. The interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a left sidebar with icons for file explorer, search, and output. The notebook content is organized into sections, each with a play button icon on the left. On the right side of the notebook, a vertical blue bar indicates the steps of Procedure 1, with numbers 1 through 9 corresponding to the sections.

- Step 1:** Input protein sequence(s), then hit Runtime -> Run all. The `query_sequence` field contains the sequence: "MAAPLVLVVAVTVRAALFRSSLAEFISERVEVVSPLSSWKRVVEGLSLDLGVSPYSGAVFHETPL". The `jobname` field contains "PIGU".
- Step 2:** Install dependencies. A link to 'Show code' is provided.
- Step 3:** MSA options (custom MSA upload, single sequence, pairing mode). The `msa_mode` is set to 'mmseqs2_uniref_env' and the `pair_mode` is set to 'unpaired_paired'.
- Step 4:** Advanced settings. The `model_type` is set to 'auto', `num_recycles` is 'auto', `recycle_early_stop_tolerance` is 'auto', `pairing_strategy` is 'greedy', `max_msa` is 'auto', `num_seeds` is '1', and `use_dropout` is unchecked.
- Step 5:** Run Prediction. The output shows a reranking of models by 'plddt' metric, with the top five models listed: rank_001 (pLDDT=94.1, pTM=0.93), rank_002 (pLDDT=93.8, pTM=0.925), rank_003 (pLDDT=93.3, pTM=0.924), rank_004 (pLDDT=92.4, pTM=0.915), and rank_005 (pLDDT=92.1, pTM=0.917).
- Step 6:** Display 3D structure. The `rank_num` is set to '1'. A 3D ribbon diagram of the protein structure is shown.
- Step 7:** Plots. A link to 'Show code' is provided.
- Step 8:** Package and download results. A link to 'Show code' is provided.
- Step 9:** Package and download results. A link to 'Show code' is provided.

Fig. 3 | Web-based ColabFold-AF2 notebook. The same notebook is used for all procedures of this protocol. The steps of Procedure 1 are indicated with their corresponding numbers. Parameters highlighted in color are explained in Box 2.

Protocol

3. Run the next cell `install_dependencies`. It does not contain any modifiable parameters.
4. Continue to `MSA options`. This cell controls the homology search for the input. Keep it at its default values and run the cell. Refer to Box 2 for alternative options for setting this cell before running it.
◆ **TROUBLESHOOTING**
5. The `Advanced Settings` cell allows for controlling the type of prediction model, recycling, MSA sampling and randomization. Use its pre-loaded default setting or set it (Box 2) and run it.
6. The next cell is `Run prediction`. Run the cell to start the prediction process with or without a preceding MSA computation step (depending on the values set in the `MSA options` cell). The five computed protein models will be ranked by their predicted local distance difference test (pLDDT) score.
◆ **TROUBLESHOOTING**
7. The next cell is `Display 3D structure`. Select the model to be displayed by its rank by using `rank_num` and then run the cell to view the result. This cell also allows for adjusting the colors and chain display.
8. Run the next cell, `Plots`, to produce the predicted aligned error (PAE), sequence coverage and pLDDT plots (discussed in Box 4 and Anticipated results).
9. Save the results as a compressed zip file by running the `Package and download results` cell.
◆ **TROUBLESHOOTING**

Predicting the structures of PIGK, PIGT, PIGS and GPAAI

● **TIMING** ~20, 30, 30 and 40 min (T4 GPU)

10. Repeat Steps 1–9 analogously for the other four monomeric units of GPIT.

Procedure 2: monomer prediction (local)

▲ **CRITICAL** Local predictions with ColabFold are carried out with a single command-line tool, `colabfold_batch`. It supports batch prediction, eliminating the need to input query sequences one by one.

Predicting the structure of PIGU

● **TIMING** ~5 min

1. Save the sequence of PIGU locally in FASTA format (`PIGU.fasta`). In addition to FASTA, ColabFold supports several types of input formats (Box 3). ColabFold can also be provided with multiple pre-computed MSAs (for MSA generation options, see Box 5) (i.e., multiple queries as a batch) by placing them within a single directory.
2. Use a single command to generate the MSA and predict the structure:

```
$ colabfold_batch Q9H490.fasta /path/to/results
```

Optionally, additional arguments can be passed to this command if non-default settings are desired. Users can refer to the complete list of parameters by running:

```
$ colabfold_batch --help
```

These parameter settings are as detailed in Box 2; however, when running locally, the user should replace underscores in the parameter name with dashes and set it by using ‘-’ before the name, for example:

```
--msa-mode mmseqs2_uniref
```

Protocol

When using pre-computed MSAs for batch prediction, use the following command to predict the structures:

```
$ colabfold_batch /path/to/input/MSA(s) /path/to/results
```

◆ TROUBLESHOOTING

3. Check the output files (detailed in Box 3) in the specified output directory (/path/to/results). Unlike web-based predictions, local predictions do not run on a notebook with 3D structure visualizations. Instead, users can explore the predicted structures in PDB format by using tools like ChimeraX and PyMOL.

Procedure 3: complex prediction (web)

▲ **CRITICAL** ColabFold's complex prediction process closely resembles that of monomers. The main differences are the input, the additional MSA pairing option and the use of a multimer prediction model. In the following steps, we focus solely on these differences, by using the GPIT complex as an example. For all other settings, which are shared with monomer prediction, we refer the reader to the monomer section of the protocol and Box 2.

Predicting the structure of the GPIT complex

● TIMING ~60 min (A100 GPU)

1. Open a web browser and navigate to <https://alphafold.colabfold.com>. Set the hardware accelerator to A100 GPU by navigating to Runtime → Change runtime type in the menu bar.
▲ **CRITICAL STEP** Because of the long length of the GPIT complex (2,500 residues), this procedure requires a paid Pro Colab account, leveraging the much larger amount of GPU RAM of the A100 GPU (40 GB of GPU RAM).
◆ **TROUBLESHOOTING**
2. Concatenate the sequences of all subunits of the GPIT complex by using a colon (':') and paste the concatenated sequence into the `query_sequence` field.
▲ **CRITICAL STEP** See Box 3 for an example concatenation in CSV format. To obtain the GPIT subunit sequences, see the Equipment section of this protocol.
3. All cells are preset to their default values, which will be used in this example. However, other options are available, and their details are provided in Box 2. These include the options available for monomer prediction and controlling the pairing process, which identifies the same taxon across the different protein subunits and the model used for protein prediction. Click `Runtime` and hit `Run all`. This action performs all prediction steps without a break.
◆ **TROUBLESHOOTING**
4. Once all cells are processed, download the various result files (Box 3) as a single compressed file, as instructed in the pop-up window. In complex prediction, the five generated protein models will be ranked by their predicted template modeling (pTM) and interface pTM (ipTM) scores (Box 4) with the following formula: $0.2 \times \text{pTM} + 0.8 \times \text{ipTM}^8$.
◆ **TROUBLESHOOTING**

Procedure 4: complex prediction (local)

Predicting the structure of the GPIT complex

● TIMING ~110 min

1. Concatenate the sequences of all subunits of the GPIT complex by using a colon (':') and save the concatenated sequence in CSV format (see the example in Box 3) as `input.csv`.

Protocol

2. Use a single command to generate the MSA and predict the structure:

```
$ colabfold_batch input.csv /path/to/results
```

◆ TROUBLESHOOTING

3. Check the results in the output directory (/path/to/results).

Procedure 5: conformation prediction (web)

▲ **CRITICAL** This procedure uses the AF2 models outside their initial scope of predicting a single conformation given an AA sequence. Here, we present two strategies for sampling different conformations, focused either on reducing MSA depth, which is controlled by ColabFold-AF2 through the `max_msa` parameter (Box 3), or on activating dropout layers by using the ASCT2 as an example. Depending on the starting point (seed), each AF2 execution can reach slightly different results. Thus, in both strategies, we increase the number of times the model is run by using different starting points, thereby increasing the chance of sampling alternative conformations. As del Alamo et al. warn²⁹, there is no one-size-fits-all approach for sampling the conformational space. We therefore recommend trying out various parameter options and, more importantly, being extra cautious when interpreting the results.

Conformation sampling of ASCT2

● **TIMING** ~ 35 min (MSA depth reduction) or 60 min (dropout) (A100 GPU)

1. Open a web browser and navigate to <https://alphafold.colabfold.com>. GPU usage is enabled by default; you can verify that by navigating to Runtime → Change runtime type in the menu bar.

▲ **CRITICAL STEP** To shorten runtimes, we recommend conducting this procedure by using an A100 GPU on a paid Pro Colab account. However, this procedure can also be performed with the default T4 GPU, taking ~7 h.

◆ TROUBLESHOOTING

2. Paste the ASCT2 sequence into the `query_sequence` field.
▲ **CRITICAL STEP** To obtain the ASCT2 sequence, see the Equipment section of this protocol.
3. Conformation prediction requires changing the default settings to increase the uncertainty of the network by modifying parameters in the Advanced settings → Sample settings field. In the MSA depth-reduction strategy, adjust the value of `max_msa` to 32:64. Alternatively, activate the dropout layers by checking the `use_dropout` box. In addition to either strategy, set `num_seeds` to its maximum (16), to generate models using different starting points (Box 2). All other cells are kept at their default values in this example. Box 2 provides information on how to modify them, if desired.
4. Click Runtime and hit Run all. This action performs all prediction steps without a break.
◆ **TROUBLESHOOTING**
5. Once all cells are processed, download the various result files (Box 3) as a single compressed file, as instructed in the pop-up window. For each seed from 0 to 15, five structure models will be computed, resulting in a total of 80 predictions ranked by their pLDDT score.
◆ **TROUBLESHOOTING**

Procedure 6: conformation prediction (local)

▲ **CRITICAL** In the command-line, instead of setting `num-seeds` (Box 2), we directly set the starting points through the `random-seed` parameter. Both these options result in a similar behavior.

Protocol

Conformation sampling of ASCT2

● TIMING ~55 min (MSA depth reduction) or 100 min (dropout)

1. Save the ASCT2 protein sequence FASTA file (`rcsb_pdb_7BCQ.fasta`) in the input file directory.
2. For the MSA depth-reduction strategy, the basic command is:

```
$ colabfold_batch rcsb_pdb_7BCQ.fasta /path/to/result --max-msa 32:64  
--random-seed z
```

For the dropout-activation strategy it is:

```
$ colabfold_batch rcsb_pdb_7BCQ.fasta /path/to/result --use-dropout  
--random-seed z
```

This basic command should be run 16 times with a different `z` value each time (0,1,...,15). To avoid typing the command multiple types, use the following bash script, which embeds the basic command in a loop. In addition, it copies the MSA computed in the first iteration to all following iterations, thereby reducing calculation time:

```
#!/bin/bash  
INPUTFILE="rcsb_pdb_7BCQ.fasta"  
OUTPUTDIR="ASCT2/32_64"  
for RANDOMSEED in `seq 0 15`; do  
  if test ${RANDOMSEED} -ne 0 ;then  
    mkdir -p ${OUTPUTDIR}/${RANDOMSEED}  
    # Copy the MSA file of iteration '0' to skip MSA computation by the  
    # webserver  
    cp -rp ${OUTPUTDIR}/0/*_env ${OUTPUTDIR}/${RANDOMSEED}  
    fi  
    colabfold_batch \  
    --random-seed ${RANDOMSEED} \  
    --max-msa 32:64 \ # Keep this line for the MSA depth reduction strategy  
    --use-dropout \ # Keep this line for the dropout-activation strategy  
    ${INPUTFILE} \  
    ${OUTPUTDIR}/${RANDOMSEED}  
done
```

◆ TROUBLESHOOTING

3. After completion, in each output directory (`$OUTPUTDIR/$RANDOMSEED`) corresponding to seed numbers 0–15, find the output files generated for the five models.

Troubleshooting

Troubleshooting advice can be found in Table 1.

Table 1 | Troubleshooting table

Step	Problem	Possible reason	Solution
Procedure 1, Step 1; Procedure 3, Step 1; Procedure 5, Step 1	Colab menu bar invisible	The menu bar is minimized	Expand the menu bar by clicking the 'v' symbol on the top right within Colab
Procedure 1, Step 6; Procedure 3, Step 3; Procedure 5, Step 4	Running out of resources. The limit of available resources provided by Colab has been reached	Analyzing longer input sequences, using many recycles or exceeding the number of times that Colab allows a user to run in a given time period	Wait (usually 1 d, although the timeout period is not fixed) before attempting to use Colab again Use a paid Pro account Run ColabFold locally

Table 1 (continued) | Troubleshooting table

Step	Problem	Possible reason	Solution
Procedure 1, Step 6; Procedure 2, Step 2; Procedure 3, Step 3; Procedure 4, Step 2; Procedure 5, Step 4; Procedure 6, Step 2	Running predictions takes too much time	GPU usage is disabled, query sequence is too long or there is a high number of seeds or recycles	Check Colab's GPU option. Set the hardware accelerator to GPU at Runtime → Change runtime type Switch to an advanced GPU for a fast response time
Procedure 1, Step 9; Procedure 3, Step 4; Procedure 5, Step 5	Results download error	Pop-ups are disabled or there are other internal Colab issues	Check pop-ups and rerun the Package and download results cell Manually download results. To do so, click on the folder icon on the left bar, navigate to the file (<jobname>.result.zip), right-click on it and click 'Download' Save to Google drive by selecting the save_to_google_drive option in the Advanced Settings cell
Procedure 1, Step 2; Procedure 2, Step 2; Procedure 3, Step 3; Procedure 4, Step 2; Procedure 5, Step 4; Procedure 6, Step 2	Custom template upload error	The template file name does not follow the PDB naming convention, or the template file is not in the proper mmCIF or PDB format	Check that the template name follows the PDB naming convention of four lowercase letters Check that the mmCIF-formatted template includes an _entity_poly_seq field. PDB-formatted templates will be automatically converted to mmCIF by ColabFold
Procedure 1, Step 4; Procedure 2, Step 2; Procedure 3, Step 3; Procedure 4, Step 2; Procedure 5, Step 4; Procedure 6, Step 2	Custom MSA upload error	The MSA file is not in the proper a3m format, or there is an error with the header format	Check that the MSA file is in the required a3m format (Box 3). In the case of complexes, make sure that the header format is correct (Box 3)

Advanced troubleshooting

Low confidence prediction

There are several possible reasons for low confidence in the predicted protein model. If the sequence coverage is low (shallow MSA), try using other sensitive homology search tools or larger environmental databases¹⁹ for constructing a deep MSA before providing it to ColabFold (Box 5). Another reason for low confidence can be the presence of disordered regions, which challenge the network. Potential disordered regions can be detected by their low pLDDT scores and then trimmed off before re-running ColabFold-AF2. A third reason can be the stochasticity of the model. In this case, changing the model type or trying a different random seed (Box 2) might help. Another mitigation is to try a different prediction network, such as ColabFold-RoseTTAFold2, because it may perform better on specific examples. Finally, in the case of complex prediction, if the whole structure receives a low ipTM score, but the individual subunits are well predicted, it can be useful to provide the well-predicted subunit structures as custom templates and re-run the complex prediction. This approach can also be beneficial when adding an interacting partner to a well-predicted complex by providing the complex model and the monomer model of the partner as custom templates.

Local predictions: CUDA version conflicts

Common sources of issues with a local installation are version conflicts and incompatibilities of the various CUDA-related drivers. We recommend using Conda and creating a new environment for the ColabFold installation. In addition, it is important not to reuse this environment for purposes other than installing ColabFold.

Local predictions: old CUDA-capable GPUs

GPUs from the Volta CUDA-capable GPU generation are the oldest GPUs that we recommend using. Setting the `--disable-unified-memory` parameter in ColabFold should allow it to run on Kepler-generation GPUs. In addition, JAX and Jaxlib libraries have to be downgraded to versions before 0.4.

Protocol

Timing

Web-based predictions were performed by using an NVIDIA T4 GPU (Procedure 1) or an NVIDIA A100 (Procedures 3 and 5). Local predictions were performed by using an NVIDIA RTX A5000.

Procedure 1: monomer prediction (web)

Steps 1–9, predicting the structure of PIGU: ~20 min

Step 10, predicting the structures of PIGK, PIGT, PIGS and GPAA1: ~20, 30, 30 and 40 min

Procedure 2: monomer prediction (local)

Steps 1–3, predicting the structure of PIGU: ~5 min

Procedure 3: complex prediction (web)

Steps 1–4, predicting the structure of the GPIT complex: ~60 min

Procedure 4: complex prediction (local)

Steps 1–3, predicting the structure of the GPIT complex: ~110 min

Procedure 5: conformation prediction (web)

Steps 1–5, conformation sampling of ASCT2: ~35 min (MSA depth reduction) or 60 min (dropout)

Procedure 6: conformation prediction (local)

Steps 1–3, conformation sampling of ASCT2: ~55 min (MSA depth reduction) or 100 min (dropout)

Anticipated results

Understanding ColabFold's results requires examining the predicted protein structures alongside their confidence scores. The results can be organized as: (i) predicted structures, (ii) computed MSA (if requested) and its quality measures and (iii) model confidence measures (Box 4). In the following sections, we give a general explanation about the interpretation of the various plots ColabFold produces. We then describe the web-based results for monomer and complex predictions and the local prediction results for conformational sampling. Although AF2 is deterministic, its use of frameworks like JAX and CUDA can lead to slight variations in prediction between runs under the same parameters because of differences in GPU models and GPU driver versions.

Using the appropriate confidence measure

ColabFold computes various confidence measures: pLDDT, PAE, pTM and ipTM (see Box 4 for an outline of their computation). Because pLDDT is a local measure, it is not sensitive to the placement of each domain in a multi-domain protein. Therefore, a high pLDDT score does not ensure high confidence in the entire structure. In the case of proteins with multiple domains, a high pLDDT alongside a low pTM score could suggest that individual domains are predicted accurately but that their relative orientation to each other is not. We therefore recommend considering pLDDT in combination with pTM. When using a complex model, ColabFold will rank the predicted protein models by the formula: $0.2 \times \text{pTM} + 0.8 \times \text{ipTM}$.

ColabFold's sequence coverage plot

The sequence coverage plot (see the example in Fig. 4a) illustrates the per-residue coverage and diversity, measured as sequence identity to the query (qid). Coverage represents the

number of homologous sequences detected per residue and is indicated as a black line in the plot. The x-axis indicates the position within the query sequence, and the y-axis indicates the MSA coverage. The upper limit of the y-axis corresponds to the MSA depth. In most cases, a minimum coverage of 30 sequences for most query residues is required for accurate prediction, preferably >100 sequences per residue¹. Next, qid indicates the sequence similarity (reflecting evolutionary distance) between each homolog and the query, where higher qid values indicate higher similarity. ColabFold encodes qid as a color, ranging from red (low) to blue (high). Each homologous sequence is illustrated as a horizontal line, on which only segments aligned with the query are visualized in color. MSAs with similar proportions of high and low qid homologs are preferable because homologs of varying evolutionary distances contribute different insights about the structure.

ColabFold's pLDDT plot and JSON file

The pLDDT scores computed for each AA are plotted by ColabFold in a single plot for the five predicted protein models (see the example in Fig. 4c). On the x-axis are the query's AAs, and on the y-axis are their pLDDT scores in each of the models (encoded in color). The pLDDT scores for each AA are additionally reported in the JSON and the PDB files produced for each model: in the JSON file under the pLDDT field and in the PDB file under the B-factor field (the second to last column). The average pLDDT score for each predicted model is provided in the output log.txt file.

ColabFold's PAE plot and JSON file

The PAE scores are computed for each of the query's residues over all other residues (Box 4) and thus can be plotted as a square in the dimensions of the query length (see the example in Fig. 4d). On the x-axis are the scored residues, on the y-axis are the points of alignment and the color reflects the PAE value (note the non-standard y-axis, with lower values on top).

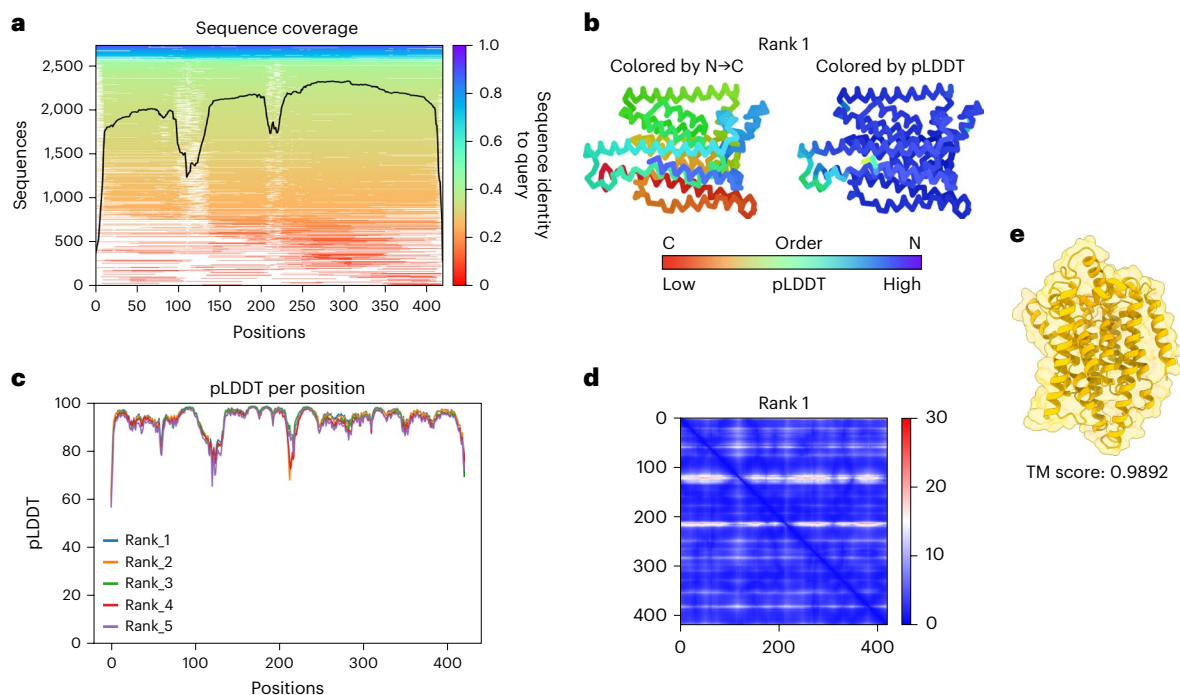


Fig. 4 | ColabFold's output for PIGU monomer prediction. **a**, Sequence coverage colored by the sequence identity to the query. The total number of homologous sequences found per residue is indicated by a black line. **b**, A 3D plot of the top-ranking model is colored by the order of residues (left) and by per-position pLDDT score (right). **c**, Per-position pLDDT scores for each of the

five predicted models. **d**, The PAE plot for the top-ranking predicted model is a nearly all-blue square, suggesting that PIGU forms a single domain. **e**, Alignment between the top-ranking predicted model (thick yellow ribbon) and the experimentally solved structure (yellow cloud).

PAE scores are not symmetric, meaning that the score at position (x,y) is generally not equal to that at (y,x). Good (low) scores are colored in blue, and bad (high) scores are colored in red. When examining a PAE plot, it is recommended to start by scanning the diagonal. Blue squares along the diagonal most likely indicate a well-predicted domain⁴⁴. See ‘Predicted aligned error tutorial’ in <https://alphafold.ebi.ac.uk/entry/Q5VSL9> for an additional example for interpreting PAE plots. The computed PAE scores used for plotting are also provided in JSON format (Box 3) in the `predicted_aligned_error_v1.json` file. This file contains two fields. The first field ‘predicted_aligned_error’ is a list (denoted by square brackets) of lists. Each internal list stores the PAE scores for each of the residues. The second field ‘max_predicted_aligned_error’ stores the highest (worst) PAE value.

Monomer prediction (PIGU)

After prediction is complete, the first result that one can see is the ‘Sequence coverage’ plot in the `Run prediction` cell. Here, we can see that the 420-AA-long query PIGU has an MSA depth of ~2,700 and a coverage that exceeds 400 homologs for every AA (Fig. 4a), which are well sufficient for structure prediction. In addition, it is well balanced, with both distant homologs (colored in red) and close homologs (blue). Next, the five protein models that ColabFold produced for PIGU appear in the cell sequentially and are ranked by their average pLDDT. Each protein model is displayed as two 3D plots, one colored by AA position (N → C) and the other by pLDDT. As can be seen in the 3D plots produced for the top-ranking model (Fig. 4b), the model consists of multiple helices arranged from the N terminus (blue) to the C terminus (red). In addition, the pLDDT scores of nearly all AAs appear in blue color, suggesting high prediction confidence. The predicted structures are additionally shown as an interactive plot in the next cell `Display 3D structure`. The next cell `Plots` provides information on the two confidence measures computed by ColabFold: pLDDT and PAE. The pLDDT plot (Fig. 4c) presents the same information visualized in the cell `Run prediction` but in greater detail. As can be seen, in all five predicted models, most of PIGU’s residues display high pLDDT scores (>70), reflecting strong local prediction reliability. Regions with slightly lower pLDDT (around residues 1, 120, 210 and 420) are likely to be linker regions or terminal. Examined together with Fig. 4a, we see these regions correspond to MSA positions with less evolutionary information (lower coverage). The PAE plot of the top-ranking model (Fig. 4d) is essentially a blue square, suggesting that PIGU is composed of a single domain. When compared to the experimentally solved structure, the top-ranking predicted model from ColabFold had a template modeling (TM) score of 0.989 (Fig. 4e), providing external support for the high average pLDDT of 94.1 (Table 2) and indicating high overall accuracy.

Complex prediction

The plots for complex prediction are similar to those of monomer prediction, with the distinction that each subunit is demarcated by a black line. The sequence coverage plot (Fig. 5a) indicates sufficient conditions for protein prediction because the resulting MSA consists of ~1,000 paired sequences as well as >1,000 unpaired sequences for each subunit, due to setting `MSA pair_mode` to `unpaired_paired` (Box 2). When analyzing the results, we focus on global confidence measures, suitable for complex prediction assessment: PAE, pTM and ipTM (Box 4). In the PAE plot for the top-ranking model (Fig. 5b), we first notice near-perfect blue squares on

Table 2 | Characteristics and confidence scores for structures predicted in Procedures 1–4

Protein	Sequence length	MSA depth	pLDDT	pTM	ipTM	TM score
PIGU	420	2,736	94.1	0.931	–	0.9892
PIGK	395	5,471	83.6	0.709	–	0.8303
PIGT	527	3,330	90.6	0.895	–	0.9216
PIGS	533	1,865	86.4	0.827	–	0.8846
GPAA1	621	2,267	86.9	0.867	–	0.9798
GPIT	2,496	18,874	87.8	0.905	0.899	0.9851

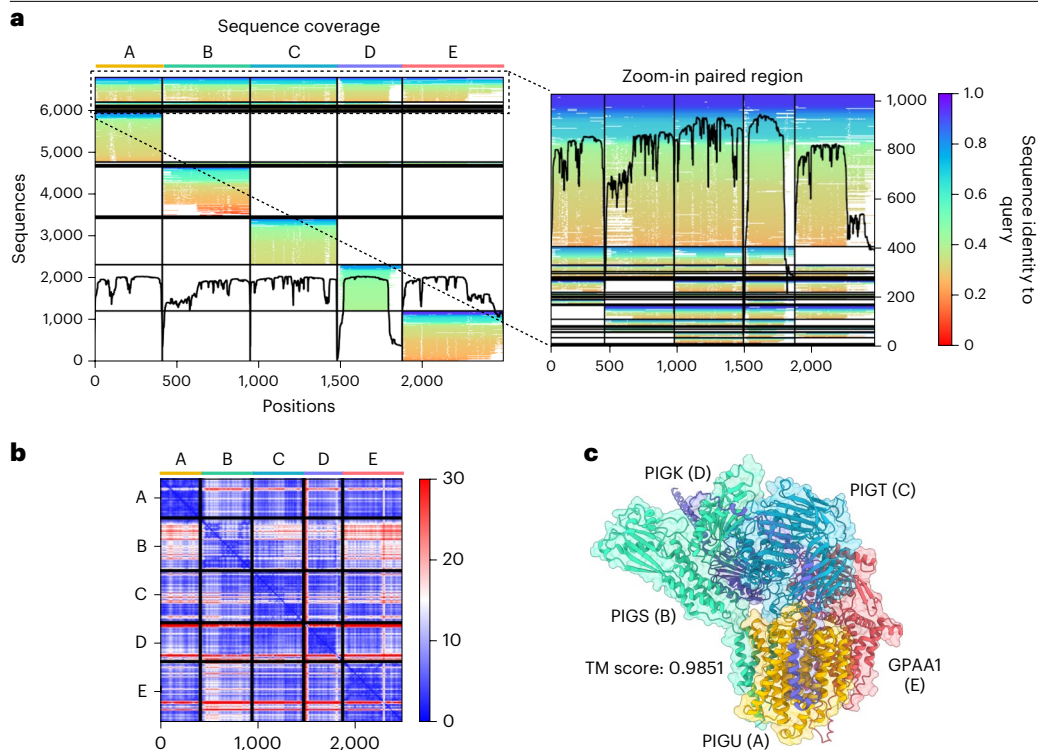


Fig. 5 | ColabFold's output for GPIT complex prediction. The five chains are demarcated with black borders or labels A–E. **a**, Sequence coverage: during pairing, ColabFold examines the homologs detected for all chains by their taxonomic assignment. For each chain, homologs could either be paired (top, magnified view) or not (bottom). In the main panel, the black chart indicates the total number of homologs per position, and in the magnified view, the total number of paired homologs. The upper section of the magnified view indicates that ~600 species had homologs for all five GPIT subunits, while the lower part represents species that had two to four homologous subunits. **b**, The PAE plot for the top-ranking model (rank_1) suggests overall confidence in the global predicted structure, despite areas of less confidence in cross-chain interplay. **c**, Alignment between the top-ranking model (thick ribbons) and the experimentally solved structure (cloud; PDB: 7W72⁴⁷). Each chain is shown in a different color.

the diagonal: one in the square of chain A and another in the square of chain D. This suggests that each of these chains forms its own globular structure, which can be predicted with confidence, relative to itself. Specifically, knowing the position of some residues in any of these chains indicates the relative positions of all others in the same chain with high accuracy. In contrast, the diagonal squares of chains C and E are crossed with red lines, suggesting that each of these chains is probably composed of two domains whose relative position to each other cannot be as confidently determined. The diagonal square of chain B is least blue, possibly indicating three short domains or an unstructured region. Next, we examine the interplay between the chains. We can see that the rows of chain B are mostly white or red for all other chains; that is, knowing the positions of residues in chain B does not allow predicting the relative positions of other chains with confidence, especially those of chain E. In contrast, the rows of chains A and D are mostly light-blue, indicating inter-chain confidence. In addition to these 'high level' insights, we notice positions of special interest. The positions of residues at the N terminus of chain D cannot be predicted with confidence, no matter where the point of alignment is, as indicated by the red columns around sequence position 1,500. In addition, all chains seem to include short segments, perhaps linkers, that are not informative for the structure of other elements, as indicated by red rows around position 100, for example. The computed pTM score for the GPIT structure is 0.9 (Table 2), suggesting high confidence in the structure, despite few lower-confidence positions. Finally, the ColabFold-AF2 top-ranking predicted model has a TM score

of 0.985 relative to the experimentally solved human GPIT structure (Fig. 5c), confirming AF2's predicted confidence (pTM) in the predicted structure.

Conformation prediction

In Procedures 5 and 6, we used two strategies to explore various conformations of ASCT2: reducing MSA depth and activating dropout layers. In addition, through 16 iterations, a total of 80 predicted protein models were generated for each of the two strategies. In Procedure 6, we produced a shared MSA for all predicted models under each strategy. As can be seen in Fig. 6a, sequence coverage is well beyond sufficient for structure prediction. Next, we found that the average value across all models of the per-residue pLDDT average (Fig. 6b) was 74.7, suggesting confidence in searching for real conformations among them. In the following, we identify alternative conformations by using principal component analysis (PCA), as described

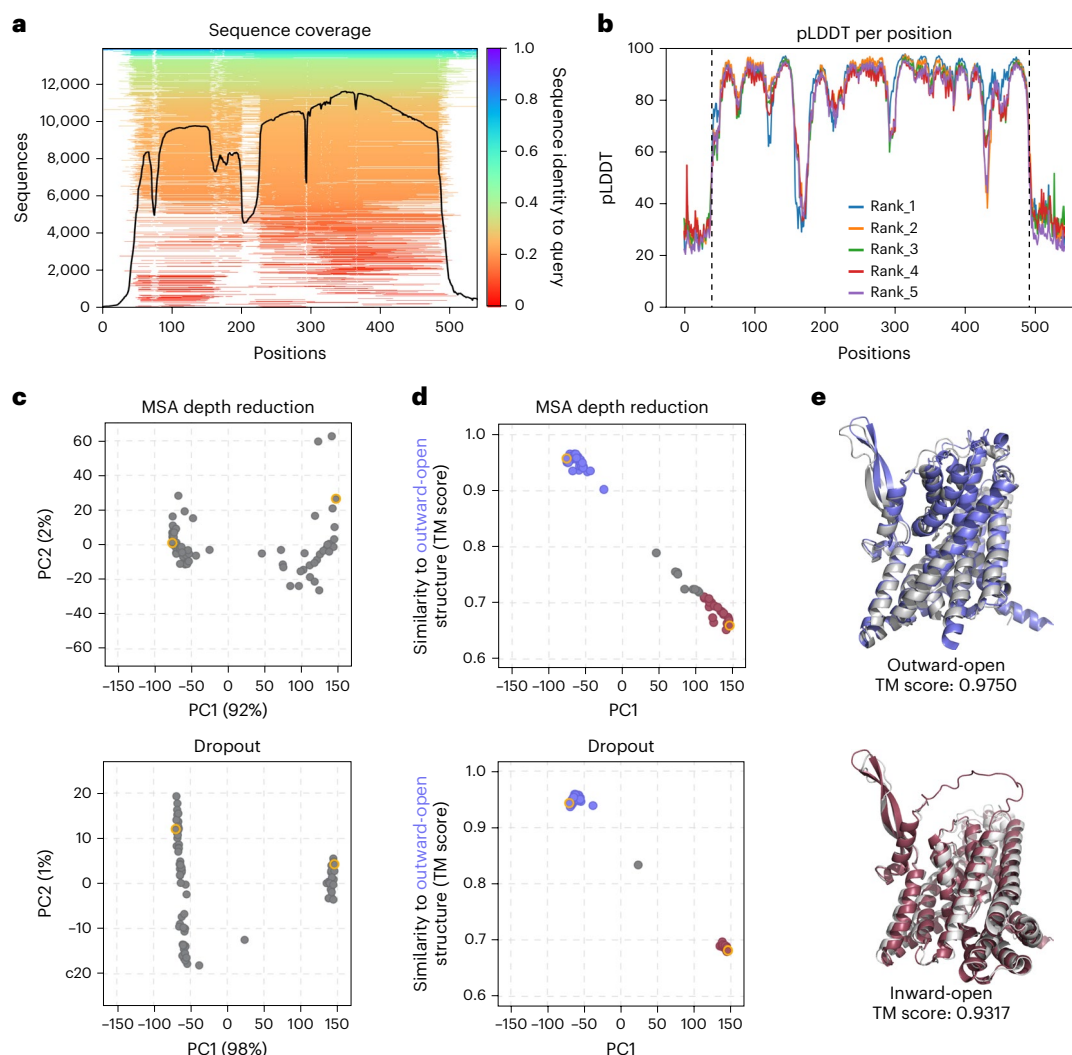


Fig. 6 | ColabFold's ASCT2 conformation prediction by MSA depth reduction or activating dropout layers. **a**, Sequence coverage is >4,000 for all residues except for the terminal regions. **b**, Representative pLDDT plot for 5 models, out of the 80, generated for each strategy. Residues 1–40 and 492–541 were identified as unstructured by their consistently poor pLDDT scores and were excluded from principal components analysis. **c**, Principal component (PC) 1 captures most of the models' variance. Representative models (orange outline) were selected at the extremes of PC1. **d**, TM scores computed between each

predicted model and the experimentally solved outward-open conformation as a function of PC1 value. High similarity ($TM > 0.85$) to the outward-open (inward-open) conformation is indicated in purple (burgundy), affirming the ability of principal components analysis to detect conformations. **e**, Structure alignment between the representative models selected from the MSA depth-reduction strategy (color) with experimentally solved outward-open (PDB: 7BCQa⁴⁸) and inward-open (PDB: 6RVXa⁴⁹) conformations (gray).

by Howe⁴⁵ and carried out by del Alamo et al.²⁹. To that end, we used a short CPPTRAJ⁴⁶ script to capture the essence of the conformational movements among the residues across the models. To reduce noise in the PCA, we trimmed off terminal stretches of ≥ 10 residues whose structure was consistently predicted as unreliable (pLDDT < 60 in $\geq 80\%$ of the models). Ideally, the first two principal components (PCs) should capture most of the variance among the generated models to enable revealing patterns of agreement and disagreement among them. This is indeed the case for ASCT2, for which PC1 is especially dominant, capturing >92% of the variance for both strategies (Fig. 6c). We therefore selected two representative models furthest apart from each other along the PC1 axis (PC1 = -76 or PC1 = 146, for the MSA depth strategy) because these potentially correspond to two conformations. In the case of ASCT2, two experimentally solved conformations are available in the PDB, allowing us to compute the similarity between each model and each conformation. Figure 6d affirms that PC1 separates the models by conformation: low PC1 values correspond to high similarity to the outward-open conformation, whereas high PC1 values correspond to low similarity to that conformation and high similarity to the inward-open conformation. Figure 6e projects the structure of the two selected representatives on the experimentally solved ASCT2 conformation that they match, indicating the high similarity between them (TM score > 0.93 in both cases).

Data availability

All sequences used in this protocol can be found in Equipment and in the PDB.

Code availability

ColabFold is available at <https://github.com/sokrypton/ColabFold> and <https://colabfold.com>. The localcolabfold installer is available at <https://github.com/YoshitakaMo/localcolabfold>. Colab prediction notebooks based on ColabFold-AF2 v1.5.3 and local prediction scripts are available at <https://github.com/steineggerlab/colabfold-protocol>, which also includes all the input and output files.

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Author contributions

G.K., S.L., E.L.K. and M.S. developed the protocol. Y.M., S.O., M.S. and M.M. developed the ColabFold software and notebooks. G.K., S.L. and H.K. performed predictions and visualized the data. S.O., M.S. and M.M. supervised the monomer and complex prediction procedures. E.L.K., Y.M., M.S. and M.M. supervised the conformation prediction procedure. G.K., S.L. and E.L.K. analyzed the results and wrote the paper, with contributions from all authors.

Competing interests

The authors declare no competing interests.

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